1. Population regression function

$$E(y|x) = E(\beta_0 + \beta_1 x + u|x)$$

$$= \beta_0 + \beta_1 x + E(u|x)$$

$$= \beta_0 + \beta_1 x$$

2. OLS estimator for simple regression model

Ordinary Least Squares (OLS) estimates

$$\hat{\beta}_1 = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{n}(x_i - \bar{x})^2}, \quad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

3. R-square

- Goodness-of-Fit

"How well does the explanatory variable explain the dependent variable?"

$$y_i = \hat{y}_i + \hat{u}_i$$

- Measures of Variation

$$SST \equiv \sum_{i=1}^{n}(y_i - \bar{y})^2 \qquad SSE \equiv \sum_{i=1}^{n}(\hat{y}_i - \bar{y})^2 \qquad SSR \equiv \sum_{i=1}^{n}\hat{u}_i{}^2$$

Total sum of squares, represents total variation in the dependent variable

Explained sum of squares, represents variation explained by regression

Residual sum of squares, represents variation not explained by regression

$$R^2 = \frac{SSE}{SST} = 1 - \frac{SSR}{SST}$$

Adjusted R-square

$$\bar{R}^2 = 1 - \frac{n-1}{n-k-1}\frac{\text{RSS}}{\text{TSS}} = 1 - \frac{n-1}{n-k-1}\frac{\sum_{i=1}^{n}\hat{u}_i^2}{\sum_{i=1}^{n}(Y_i - \bar{Y})^2}$$

4. OLS estimator for multiple regression model

$$\hat{\beta}_1 = \frac{\sum_{i=1}^{n}(\hat{r}_{i1} y_i)}{\sum_{i=1}^{n}(\hat{r}_{i1})^2}$$

5. Variance of estimator

$$Var(\hat{\beta}_j) = \frac{\sigma^2}{SST_j(1 - R_j^2)},$$

Estimating the error variance

$$\hat{\sigma}^2 = \left(\sum_{i=1}^{n}\hat{u}_i^2\right) / [n - k - 1]$$

6. Critical value for normal distribution:

   - If the number of degrees of freedom is large enough so that the normal approximation applies, the following rules of thumb apply:

$\|t-stat.\| > 1.645$	⟶ "statistically significant at 10 % level"
$\|t-stat.\| > 1.96$	⟶ "statistically significant at 5 % level"
$\|t-stat.\| > 2.576$	⟶ "statistically significant at 1 % level"

7. F-test

   - **Test statistic**

Numerator degrees of freedom $= df_r - df_{ur}$

$$F = \frac{(SSR_r - SSR_{ur})/q}{SSR_{ur}/(n-k-1)} \sim F_{q,n-k-1}$$

The relative increase of the sum of squared residuals when going from H_0 to H_1 follows a F-distribution (if the null hypothesis H_0 is correct)

Denominator degrees of freedom $= df_{ur}$

8. LLN

Law of Large Numbers (LLN): If $Y_1, \ldots, Y_n$ are i.i.d. and $\sigma_Y^2 < \infty$, then $\bar{Y}$ is a consistent estimator of μ_Y, that is, for any $\epsilon > 0$,

$$P(|\bar{Y} - \mu_Y| < \epsilon) \to 1 \text{ as } n \to \infty .$$

It is written as $\bar{Y} \xrightarrow{p} \mu_Y$, for which we can read "$\bar{Y}$ converges in probability to μ_Y".

9. CLM

   - **Classical linear model assumptions in MLR:**
     - MLR.1: $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \ldots + \beta_k x_k + u$
     - MLR.2: random sampling from the population
     - MLR.3: no perfect collinearity in the sample
     - MLR.4: $E(u|x_1, \ldots, x_k) = E(u) = 0$ (exogenous explanatory variables)
     - MLR.5: $Var(u|x_1, \ldots, x_k) = Var(u) = \sigma^2$ (homoskedasticity)
     - MLR.6: the distribution of u given $(x_1, \ldots, x_k)$ is $Normal(0, \sigma^2)$

10. Functional Form

    - **Summay of Functional Forms involving Logarithms**

Model	Dep. Var.	Indep. Var.	β_1
Level-Level	y	x	$\Delta y = \beta_1 \Delta x$
Level-Log	y	$\log(x)$	$\Delta y = (\beta_1/100)\%\Delta x$
Log-Level	$\log(y)$	x	$\%\Delta y = (100\beta_1)\Delta x$
Log-Log	$\log(y)$	$\log(x)$	$\%\Delta y = \beta_1 \%\Delta x$